



Robotics Marketers Pty Ltd

Feasibility Study

< AI-Powered Web Application for Bot Listening >

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1. Introduction

The purpose of this project is to develop an AI-Powered application for topic-based community and organization that rely on heavy research. This application will also be useful for research analysts that spend a lot of time in research. On successful implementation, we will be able to address inefficiencies in manual research while also being scalable to different industries and topics. This tool can also be applied to any sector, from startups to non-profits.

1.1. Problem Identification

On average, a knowledge gathering worker spends more than two hours a day gathering information according to [1]. This is a very time consuming task and it highlights the issue of efficiency with manual searches. Resource gathering for a particular topic is very time consuming. The work however doesn't end at resource gathering. The resources thus gathered need to be ranked according to relevance. Without the use of any AI powered relevance scoring system, the entire resource gathering process can get very time consuming.

This project aims to benefit organizations and employees who spend a lot of time researching. The primary stakeholders for this project are knowledge gathering workers who spend about two hours on average daily to research [1]. This would include research analysts and potentially students who need to gather articles for their research timely and efficiently. Their early adoption will directly influence the success of the project. Research based organisations can expect productivity gains.

Secondary stakeholders include IT teams, legal teams and API providers. IT teams will ensure seamless operation of the application, minimizing downtime and improving operations with time. The nature of the project requires the application to pull resources from other resources. These resources can be the intellectual property of other organizations/individuals. The IT team will likely need to work closely with the legal team to ensure the resources are pulled legally and does not violate any intellectual property rights.

Tertiary stakeholders will include investors, marketing team and end users. Even though they are tertiary stakeholders, ensuring their feedback is taken into account will be instrumental in determining the project's success.

1.2. Opportunities

The application provides the opportunity to Robotics Marketer to tap into a large audience of research analysts which was previously not available to the organisation. Having access to a new audience will allow the organization to grow their knowledge base and potentially make a product accessible to other untapped markets. This will also help the organization grow its knowledge base, which is a key for success and open other opportunities in the future.

By launching an initial MVP in 6-9 months, the company can generate subscription revenues early. Other sources of revenue can be advertisements displayed on the application. The MVP can be pitched to businesses to provide a customized tool, tailored to their needs.

The projected ROI for enterprises will be about 12-18 months through time savings and enhanced research productivity.

With the tool being something Robotics marketer can claim as its own, they establish themselves as innovators, who provide AI driven solutions. This opens the door to competitive connections. The nature of the application is very scalable, which allows expansion into multiple topics and industry sectors.

Based on the knowledge base Robotic Marketer Pty Ltd will gather down the line, trend radar can be implemented in the application. This technology aims at detecting patterns in the topics researched by an individual and suggest the analyst similar topics before they actually look it up. This would save the analyst precious time the system would take in processing the request or the user would take in coming up with an appropriate prompt.

1.3. Mandates

The project must meet several statutory and organisational requirements. It must comply with privacy regulation and relevant government laws, including the Australian Privacy Act 1988, as well as international data protection standards such as GDPR and CCPA to ensure lawful data storage and processing. The AI models must adhere to ethical guidelines. Measures should be taken to remove bias from the training dataset. Any Machine language model or datasets used should have appropriate licensing for commercial use to avoid violation of intellectual property rights.

Other mandates include adherence to API usage policies from source platforms, implementation of front end security measures such as securing API keys and sanitizing untrusted input. Measures to include user tracking to track their activities to track unusual activities. The system must also meet WCAG 2.1 AA accessibility standard to ensure inclusive usability across all audiences.

1.4. Assumptions

The most important assumption for the success of this project is the availability of public API and datasets for data retrieval. The project also requires access to high quality data for model testing and tuning. It is also assumed that cloud based computing resources, such as AWS and Azure, will be available for AI model development. Mongo DB will be made accessible to store huge datasets to reproduce results and other purposes.

From a technical standpoint, Django will be the backend framework with access to basic NLP libraries. Branding guidelines will be provided early on to ensure a consistent user interface and experience. The target browsers will include latest versions of Chrome, Edge and Safari on both desktop and mobile and the system will temporarily cache responses where appropriate.

From a risk perspective, it is assumed that API access will remain available at current rate limits and pricing and that no major policy change will occur that would impact data retrieval.

2. Current Situation

There is a clear gap in the market for a tool capable of aggregating and ranking individuals and organizations by topic relevance across platforms. The creation of such a tool would enable Robotic Marketer Pty Ltd to provide their clients with highly relevant information on current conversations regarding their specific industry and location. Additionally, no system exists to automate semantic similarity matching between user queries and online conversations, reducing the precision and scalability of the company's operations. The current reliance on more manual methods is highly inefficient, due to the time-consuming nature, leading to the increased difficulty to scale. Furthermore, the lack of tools being available that provides results that are specific to the clients' industry and location, can lead to difficulty identifying the highly relevant data that is needed in order to create the appropriate marketing strategies for clients.

3. Benefits

The proposed AI-driven lead discovery and market intelligence platform will provide both tangible and intangible benefits, enabling measurable cost and time savings while also delivering strategic advantages in client relationships, brand positioning, and operational efficiency.

3.1. Tangible Benefits

Cost Savings

- Reduce labour costs by cutting manual research hours by up to **60%**.
- Lower marketing expenditure by **20–30%** through higher-precision targeting and reduced ad spend wastage.
- Increase revenue by **15–20%** via improved conversion rates from higher-quality leads.

Time Savings

- Reclaim **3–5 staff hours per week** per researcher by automating lead discovery.
- Accelerate decision-making cycles by **40%** through instant AI-driven insight generation.
- Increase lead generation throughput by **50%**, enabling faster campaign rollouts.

New Products and Services

- Launch premium AI Discovery Bot features with potential **15% uplift** in upsell revenue.
- Offer personalised AI-driven intelligence reports as an added-value service to top-tier clients.
- Enhance lead targeting accuracy via semantic relevance scoring, directly benefiting marketing, sales, and research teams.

3.2. Intangible Benefits

- **Increased client trust** through transparent, data-backed AI insights, improving credibility and authority.
- **Brand differentiation** as a modern, intelligence-led discovery platform, positioning the organisation ahead of competitors.
- **Improved inter-departmental collaboration** through consistent, standardised data formats across teams.
- **Higher employee morale** by removing repetitive, low-value manual work, allowing staff to focus on strategic and creative tasks.
- **Strengthened client confidence** in the organisation's ability to deliver accurate, timely, and relevant market insights.

4. Alternative Solutions

4.1. Alternative 1 – In-House AI-Powered Web Application

Developing an AI-powered web application entirely in-house would give Robotic Marketer complete ownership over the design, functionality and underlying technology. This approach would allow the team to craft a solution specifically tailored to client requirements while ensuring seamless integration with the company's existing AI platform. By keeping development internal, the business retains full control over data usage, privacy protocols and system architecture, which can be crucial for maintaining competitive advantage.

4.1.1. Advantages

- **Scalability and adaptability:** Can evolve with client needs and market trends. *Example:* Canva's in-house branding kit allowed them to respond rapidly to enterprise feature requests without third-party delays.
- **Full control over design, functionality, and data usage:** Enables unique features and ensures compliance with privacy/security standards.
- **Seamless integration with Robotic Marketer's existing AI platform:** Allows for a unified user experience and efficient data flow.
- **Future innovation potential:** Could integrate with voice assistants or chat-based research tools for hands-free, conversational insights (similar to Bloomberg Terminal's voice search).

- **Continuous improvement through machine learning feedback loops:** AI models can refine outputs over time based on user interactions, much like Netflix's recommendation system.

4.1.2. Disadvantages

- **High upfront and ongoing costs:** Developing an in-house AI-powered platform would require a substantial initial investment in both time and resources, as well as ongoing expenses for infrastructure, updates and maintenance. The overall financial commitment could be significant compared to a third party solution.
- **Potential resource strain:** May divert focus from other high-priority product features in the pipeline.
- **Longer testing cycles:** AI features often require iterative testing, model retraining and tuning, which can extend quality assurance (QA) timelines compared to non-AI systems.
- **Ongoing dependency on third-party APIs:** Even with an in-house build, real-time data sources may still rely on external providers whose stability is outside company control.

4.2 Alternative 2 - No Action

4.2.1. Advantages

A no action decision would imply that the organization would not have any initial financial expenditure in terms of the development or acquisition of an AI-powered web application. It could be especially advantageous in case it is necessary to address the issue of underfunding because all the funds available could be spent on other business needs. The company also evades risks that could arise as a result of AI implementation including system and security breaches and integration failures by not developing new technology. Also, there are no requirements of technical support on a continuous basis, software updates, as well as specific maintenance teams to take care of since it comes without a new AI-powered system to manage. This will enable the company to wholly focus on streamlining and enhancing the available features of its AI marketing strategy so that current equipment and procedures can perform to their best capacity without being distracted by other novel untested projects.

Bulletpoint version:

- No financial investment
- Avoids technology risks
- No maintenance burden
- Allows focus on existing AI marketing features

4.2.2. Disadvantages

Although the no action option is less expensive and less risky, it has major disadvantages. Without AI driven web applications, data will still be collected manually which is commonly slower, resource demanding and subject to errors involving the idle hand. This may hamper the speed with which the organization can determine the trends, consumer habits, or market changes. Also, the company might lose to the extent of growth that competitors can achieve because they pursue AI solutions and integration that enables them to automate, use predictive analytics, and become more engaging to their customer base. This may eventually result in a competitive disadvantage in the long run since competing organizations using AI and machine learning will have quicker access to insights, perform more efficiently, and be better equipped to act on the needs of the market. After all, failure to do anything might lead to the firm being left behind in a world that is becoming more and more technology oriented.

Bullerpoint version:

- Continued manual data collection
- Missed growth opportunities
- Falling behind competitors adopting AI-powered tools

4.3. Alternative 3 – White-Label AI Integration

In this option, the development team would integrate and customise a commercially available white-label AI lead generation and market intelligence platform. These platforms typically come pre-built with core capabilities such as data aggregation, trend detection, predictive analytics, and dashboard visualisation, while still allowing for branding, workflow tailoring, and API extensions to suit specific organisational needs.

4.3.1. Advantages

- **Faster Deployment:** Potentially functional within **8–12 weeks**, significantly faster than a full in-house build.
- **Lower Upfront Costs:** Subscription/licensing model reduces initial capital expenditure.
- **Scalability:** Flexible resource scaling as demand grows.
- **Mature Core Features:** Access to ready-made, tested functionalities (data aggregation, analytics, visualisation).
- **Customisation Potential:** Ability to add branding, adjust workflows, and integrate with other systems via APIs.
- **Regulatory Compliance:** Vendor solutions can be selected based on adherence to GDPR, CCPA, and commercial-use licensing.

4.3.2. Disadvantages

- **Ongoing Costs:** Recurring subscription or licensing fees may be substantial over time.
- **Vendor Dependence:** Reliance on a third-party provider for core infrastructure and updates.
- **Customisation Limits:** May not allow deep, highly tailored changes to underlying architecture.
- **Data Security Risks:** Sensitive data handling depends on vendor's security measures.
- **Feature Constraints:** Must operate within the vendor's feature set unless they offer extended API capabilities.

5. Recommended Solution

5.1 Decision

Continue to develop the web application that provides AI-driven discovery in-house. Ensure that this application interfaces with all Robotic Marketer systems. This way, we have full control over data, its ranking, UI/UX, and are able to serve the "quick change" demands of our sponsors.

5.2 Key Reasons Why

Compliance and control: Simpler to satisfy all the mandates for privacy, API ToS, and accessibility.

Fitting and Scalability: Customized for the exact use cases of Robotic Marketer; adding connectors and improving rankings is now even easier.

5.3 Implementation Road Map Plan

Phase 1 – Discovery & Guardrails (Weeks 1–3 → D1: 14 Aug)

Deliver the Team Manual and the Feasibility Report. Secure the validation of the minimal viable product's scope, the success metrics, and the privacy and application programming interface guardrails with the sponsor. Provide a listing of the essential data sources and connections. Schedule regular consistent sponsorship meetings for every academic term.

Phase 2 – MVP Build (Weeks 4–7 → D3: 2 Oct)

Create the workflow: data fed in (topic/area of interest/sector)

→ outputs that are organized and ranked (in terms of importance)

→ and with some basic hedging, too.

That central flow should be buildable in the next couple of weeks. I envisage a simple prototype working with version 1 of the ranking method: keyword/TF-IDF + embeddings. As for the data storage issue: I'd prefer MongoDB Atlas; its beginner friendly, it reduces operational overhead, and it has all the necessary backend features to oversee rate constraints and implement caching. Basic checks for AA accessibility and some invoice-level tests for unit/integration tests. Produce a simple working prototype that is demonstrable.

Phase 3 – Beta Refinement (Weeks 8–10 → D4: 25 Oct)

Incorporate a relevance feedback loop, deduplication, and entity enrichment. UI/UX: break content apart; remember search criteria; make errors mean something. Observe users. Observe then do some simple analysis of what you've seen. Prepare the manual. Think user; think training. Fill in the gaps: privacy, accessibility, and compliance.

Phase 4 – Demo & Hardening (Weeks 11–12 → D5: 6 Nov)

Correcting errors and enhancing speed. End to end demo scenario linked to success metrics. The final stages are to ensure the system is safe and easy to operate.

Phase 5 – Handover & Stretch (Weeks 13–14 → D7: Nov 6–20)

Provide a handover of all necessary documents in an orderly manner; include runbooks and configuration templates. Omit any and all confidential documents. Meeting to transfer sponsorship. Write down all the things we didn't get to that are unprioritized and are potential future enhancements. If time allows, make stretch features optional.

5.4 Key Success Metrics

- Reduced search time: 50–60% less time spent searching than the usual manual approach.
- Relevant hit rate:
- Target a relevance hit rate of 30% or higher for top-10 returned items that are accepted as relevant.
- Time-to-first-useful-result: 10 seconds or less with normal operation.
- Compliance and quality: No critical violations of privacy or APIs; 95% or more of automated AA checks pass.
- Adoption: 5 or more internal pilot users; 2 or more sponsored pilot use-case implementations by Beta.

5.5 Risk and Mitigations

- API volatility: caching, provider redundancy, small curated fallback dataset.
- Licensing/commercial use: maintain a register of models/datasets and licenses from the start.
- Scope creep: lock MVP at Phase 1; stretch items deferred to Phase 4.
- Student capacity/time: weekly meetings with the sponsor; maintain a small but prioritized set of tasks for the team.

REFERENCES:

[1] S. Probst, "Reality Check: Still Spending More Time Gathering Instead of Analyzing," Forbes Technology Council, Dec. 17, 2019. [Online]. Available: <https://www.forbes.com/councils/forbestechcouncil/2019/12/17/reality-check-still-spending-more-time-gathering-instead-of-analyzing/>